SolarTrack: Adaptive Solar Panel Orientation

Abstract

SolarTrack adjusts panel tilt every fifteen minutes using a lightweight sun-position model, increasing daily energy capture by twenty eight percent compared to fixed-angle panels.

Introduction

Fixed-angle solar panels lose significant output as the sun moves across the sky. SolarTrack addresses this by recomputing an optimal tilt angle at a regular interval throughout the day.

Method

SolarTrack uses a Kalman filter to smooth noisy light-sensor readings before computing the optimal tilt angle for the next fifteen-minute window.

Results

In field trials across four rooftop installations over three months, SolarTrack consistently outperformed fixed panels, with the largest gain of thirty five percent observed in winter months.

Conclusion

SolarTrack demonstrates that lightweight, sensor-driven tilt adjustment can meaningfully improve solar capture without complex hardware.